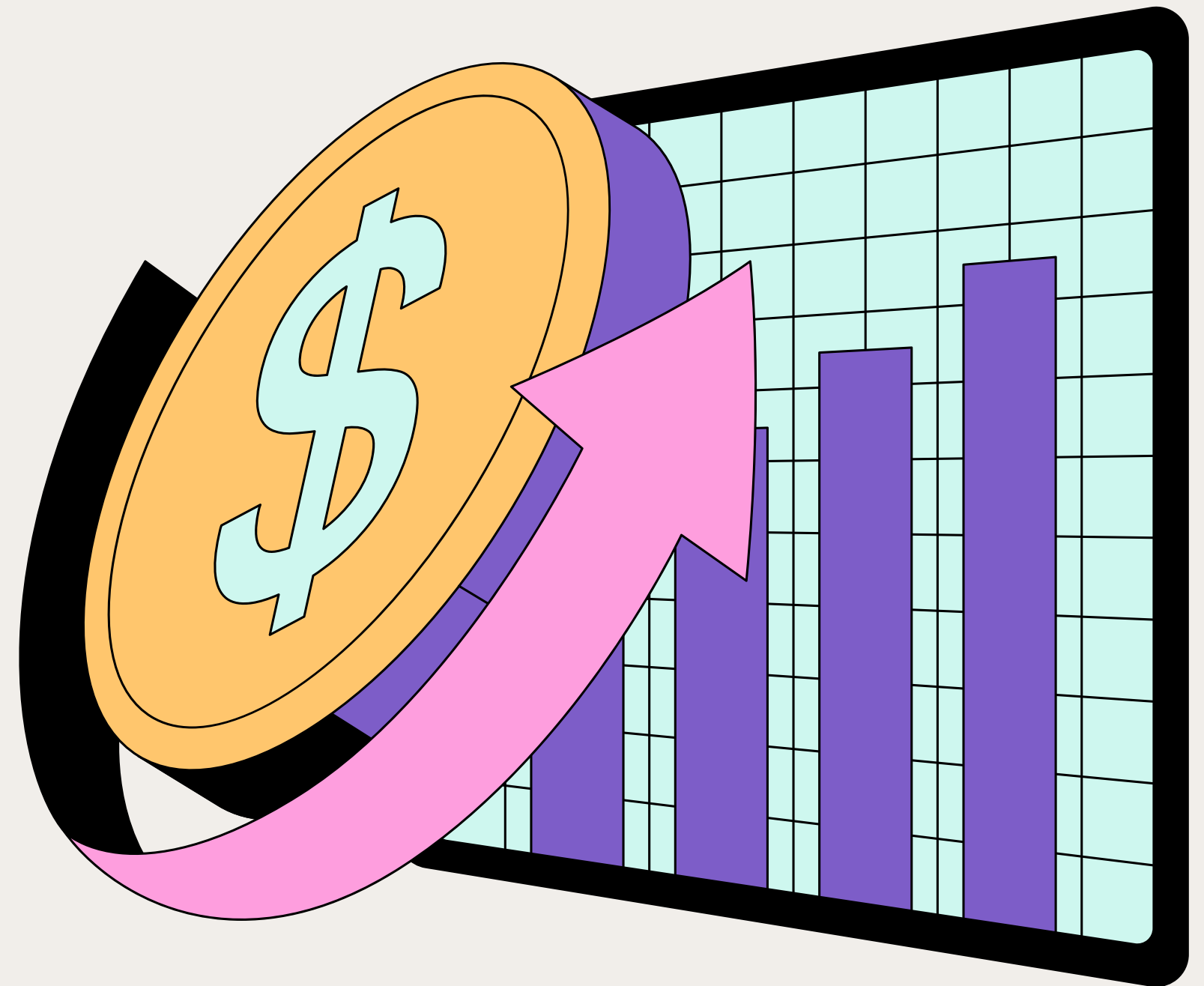


Super Store

Optimalisasi Profit dengan Machine Learning

Marcella Aurelia Yatijan - 71477



Tujuan

Memprediksi Profit dan Rugi
Order Berdasarkan Pola
Historis untuk Optimasi
Keputusan Bisnis

Mengetahui variabel yang
paling berpengaruh
terhadap profit



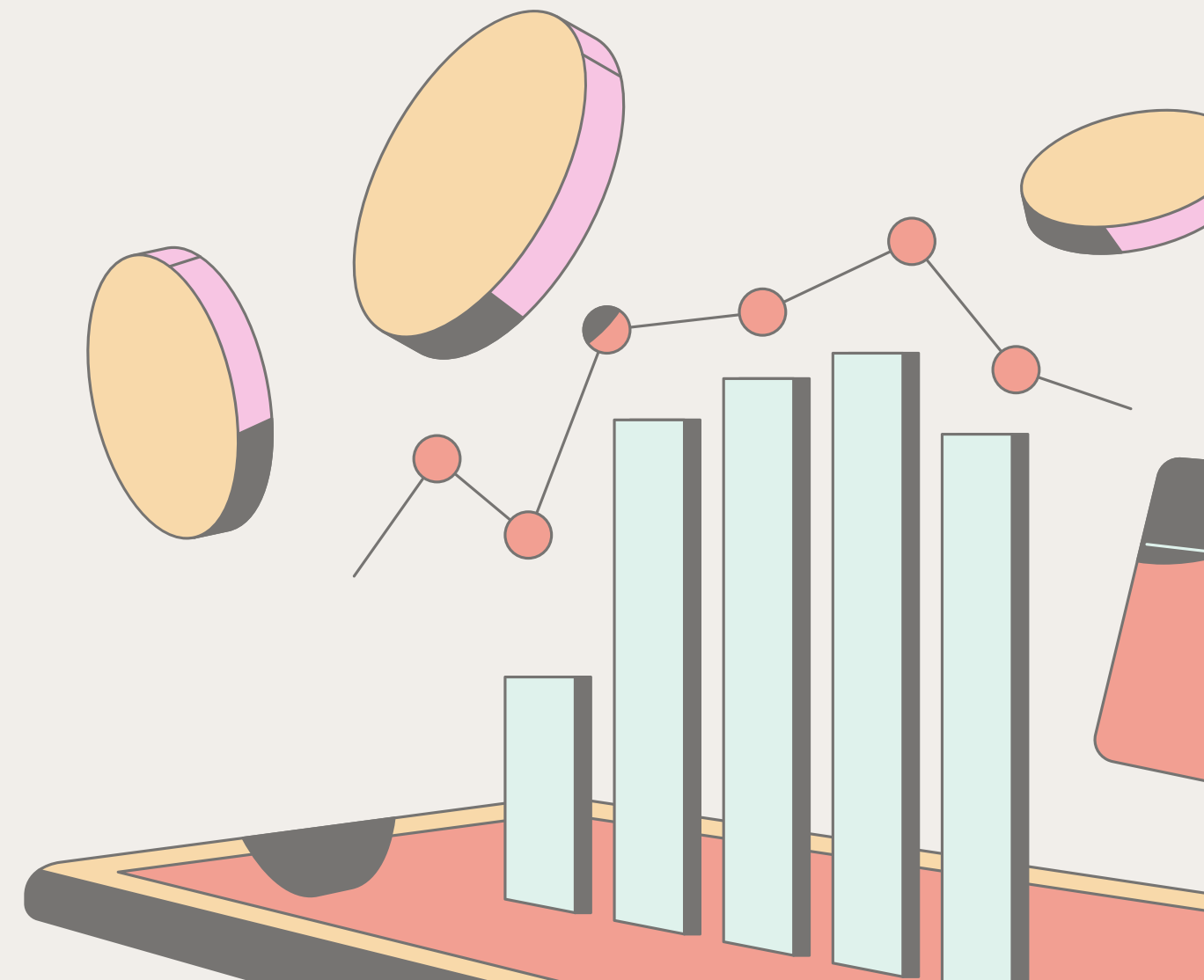
Dataset

FEATURE

- DISCOUNT
- DISCOUNT_BIN
- SALES
- CATEGORY
- MARKET
- SUB_CATEGORY
- ORDER_PRIORITY
- SEGMENT
- SHIPPING_COST
- PRODUCT_NAME
- COUNTRY
- REGION
- QUANTITY
- SHIP_DATE
- CUSTOMER_NAME
- SHIP_MODE

TARGET

- PROFIT_BIN



Profit dan Discount Bin

Kelas	Syarat	Penjelasan
'Rugi'	$p < 0$	Jika profit negatif (rugi)
'Untung Kecil'	$0 \leq p \leq 30$	Jika profit sedikit
'Untung Besar'	$p > 30$	Jika profit besar

=== Statistik Profit ===

Mean : 28.64

Median : 9.24

Modus : [0.0]

discount_bin	Range Diskon (discount)	Penjelasan
0	0.00	Tanpa diskon
1	>0.00 s.d. 0.10	Diskon rendah (0–10%)
2	>0.10 s.d. 0.30	Diskon sedang (10–30%)
3	>0.30 s.d. max_val (mungkin 0.8–1.0)	Diskon tinggi (>30%)



Feature Importance by Chi Squared

===== Hasil Uji Chi-Squared terhadap 'profit_bin' =====					
Feature	Chi2 Hitung	Chi2 Tabel	p-value	Kesimpulan	
discount_bin	36065.6884	12.5916	0.0000	✓	Tolak H0
product_name	28422.4752	7777.5746	0.0000	✓	Tolak H0
country	24149.4983	332.8538	0.0000	✓	Tolak H0
sub_category	10604.9665	46.1943	0.0000	✓	Tolak H0
category	6220.4128	9.4877	0.0000	✓	Tolak H0
ship_date	4812.2120	3052.9553	0.0000	✓	Tolak H0
region	3323.9449	36.4150	0.0000	✓	Tolak H0
customer_name	2643.4157	1681.8204	0.0000	✓	Tolak H0
market	1397.6546	21.0261	0.0000	✓	Tolak H0
order_priority	4.2271	12.5916	0.6460	Gagal	Tolak H0
ship_mode	3.1470	12.5916	0.7902	Gagal	Tolak H0
segment	1.0461	9.4877	0.9027	Gagal	Tolak H0



FEATURE

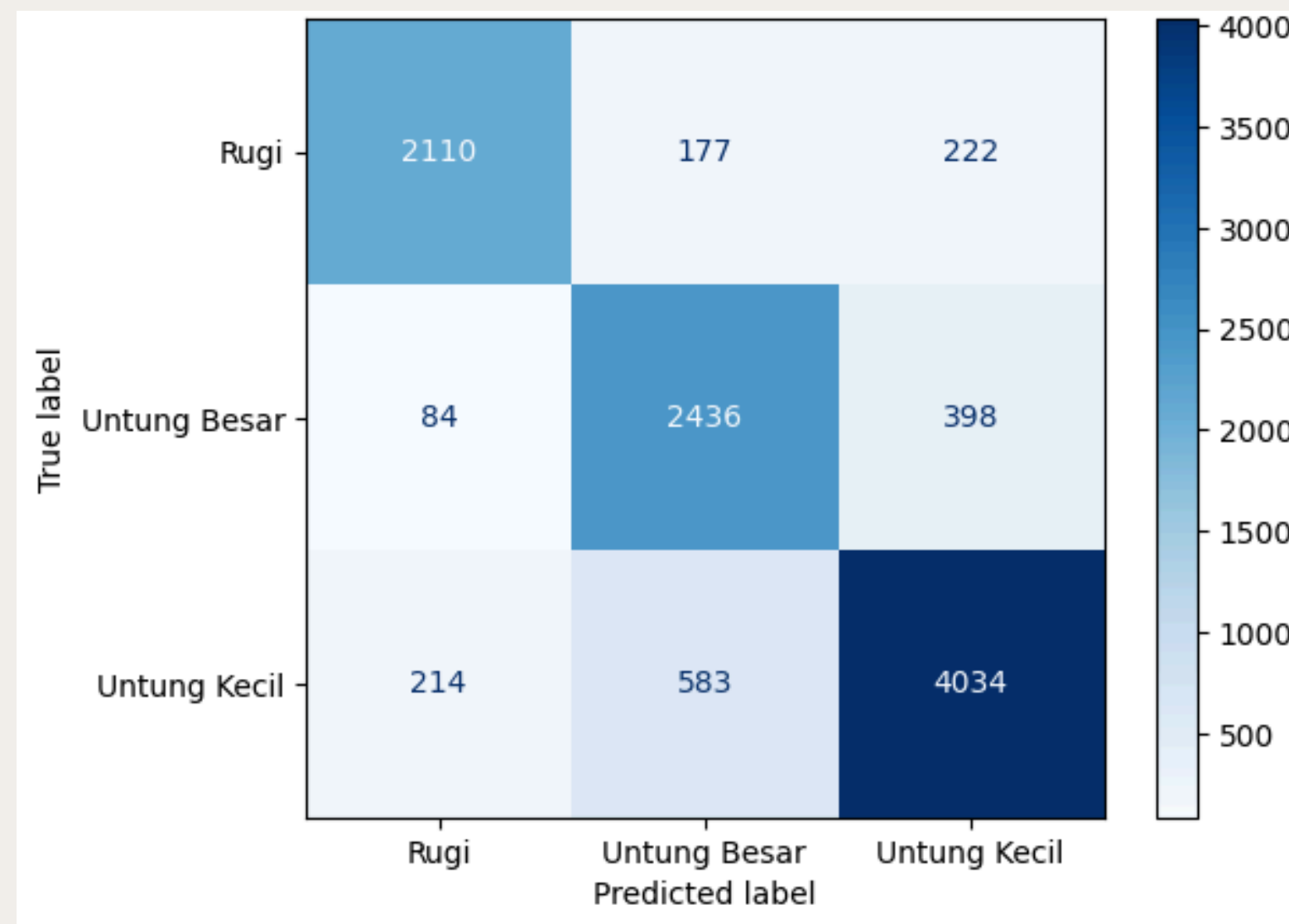
- DISCOUNT
- SALES
- CATEGORY
- MARKET
- SUB_CATEGORY
- SHIPPING_COST
- PRODUCT_NAME
- COUNTRY
- REGION
- QUANTITY
- SHIP_DATE
- CUSTOMER_NAME

Model Accuracy

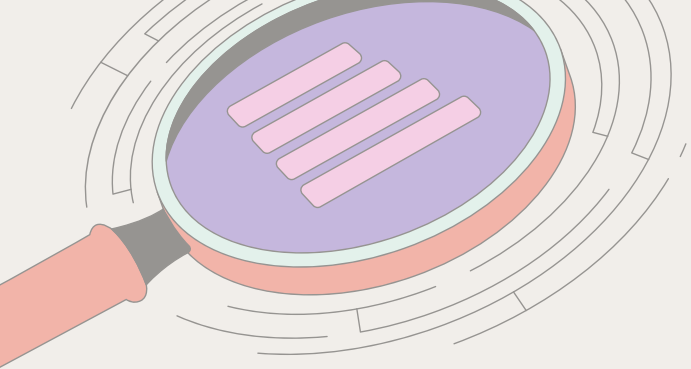
Model	Accuracy
Random Forest	84%
XGBoost + Optuna	84%
XGBoost	83%
StackingClassifier	83.4%
BaggingClassifier	82%
GradientBoostingClassifier	82%
VotingClassifier	77%
KNN	58%



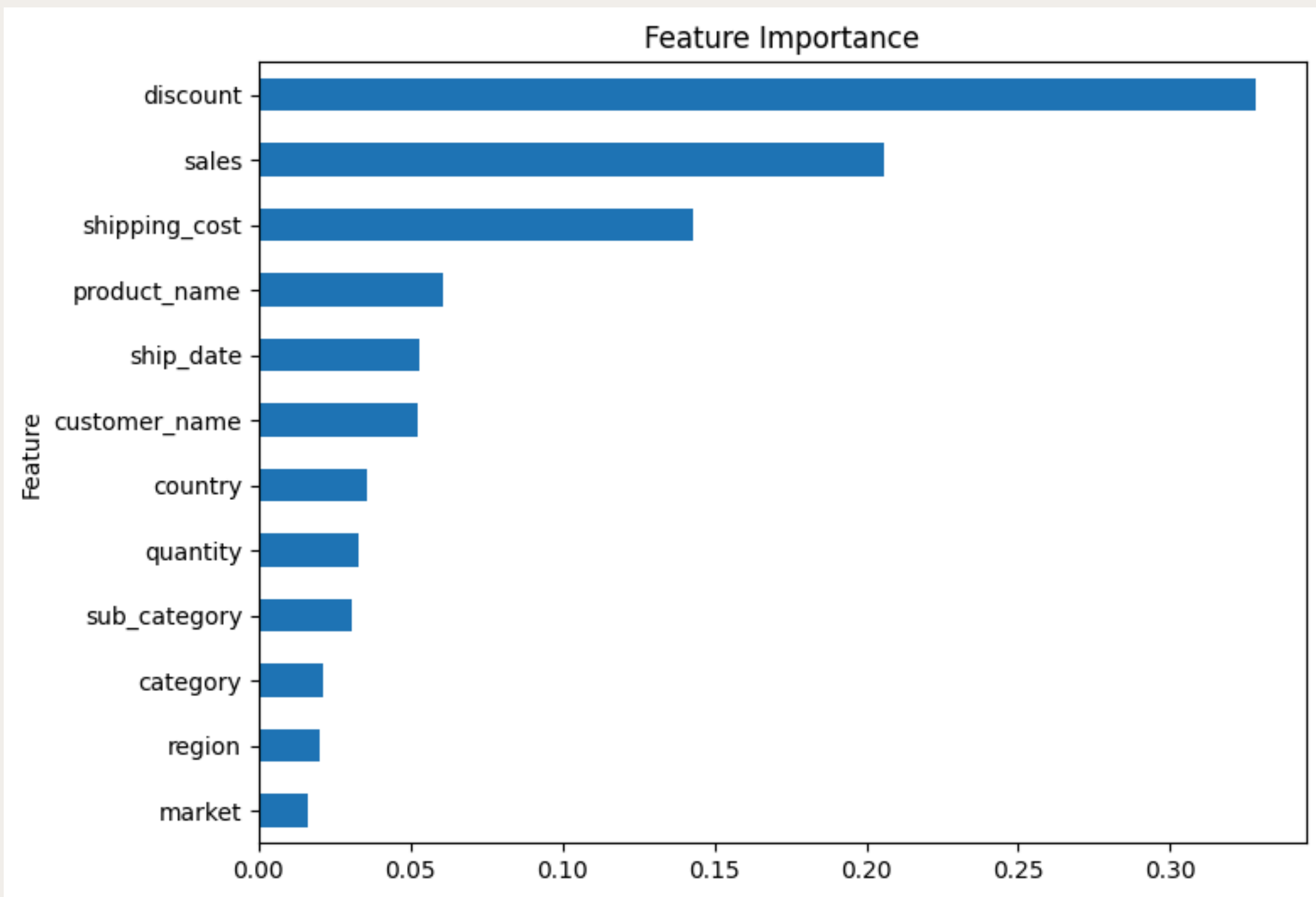
Random Forest



ACCURACY 84%
ROC AUC SCORE (OVR) 0.9562



Feature Importance by Random Forest

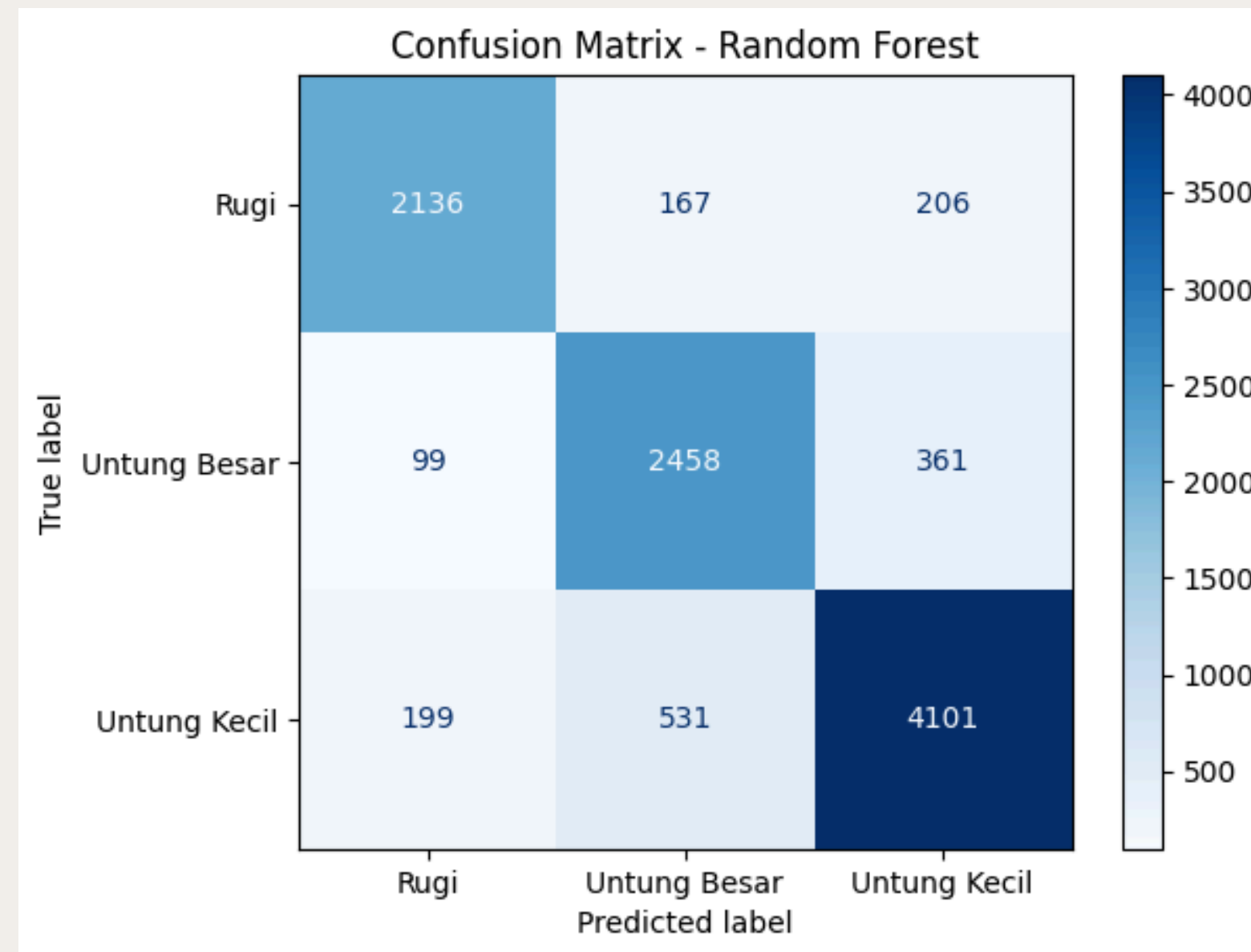


Feature	Chi2 Hitung
discount_bin	36065.6884
product_name	28422.4752
country	24149.4983
sub_category	10604.9665
category	6220.4128
ship_date	4812.2120
region	3323.9449
customer_name	2643.4157
market	1397.6546
order_priority	4.2271
ship_mode	3.1470
segment	1.0461

Model dengan 6 Feature

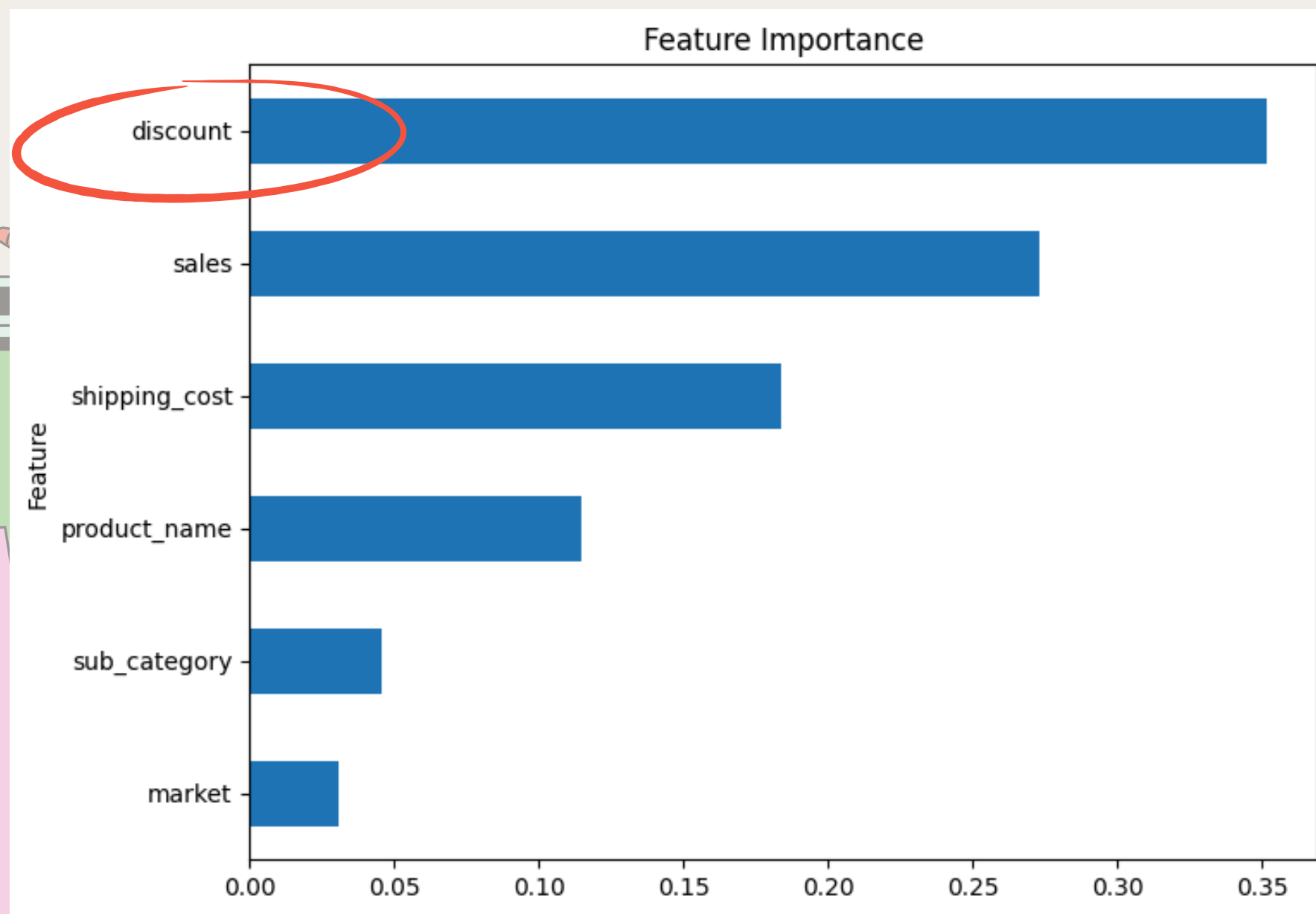
FEATURE

- DISCOUNT
- SALES
- MARKET
- SUB_CATEGORY
- SHIPPING_COST
- PRODUCT_NAME
- ~~COUNTRY~~
- ~~REGION~~
- ~~QUANTITY~~
- ~~SHIP_DATE~~
- ~~CUSTOMER_NAME~~
- ~~CATEGORY~~



ACCURACY 85%
ROC AUC SCORE (OVR) 0.9616

Model dengan 6 Feature



```
=== Feature Importance ===  
      Feature  RF Importance  
0  discount  0.351595  
1   sales    0.272951  
2 shipping_cost 0.184113  
3 product_name 0.115046  
4 sub_category 0.045771  
5   market    0.030524
```

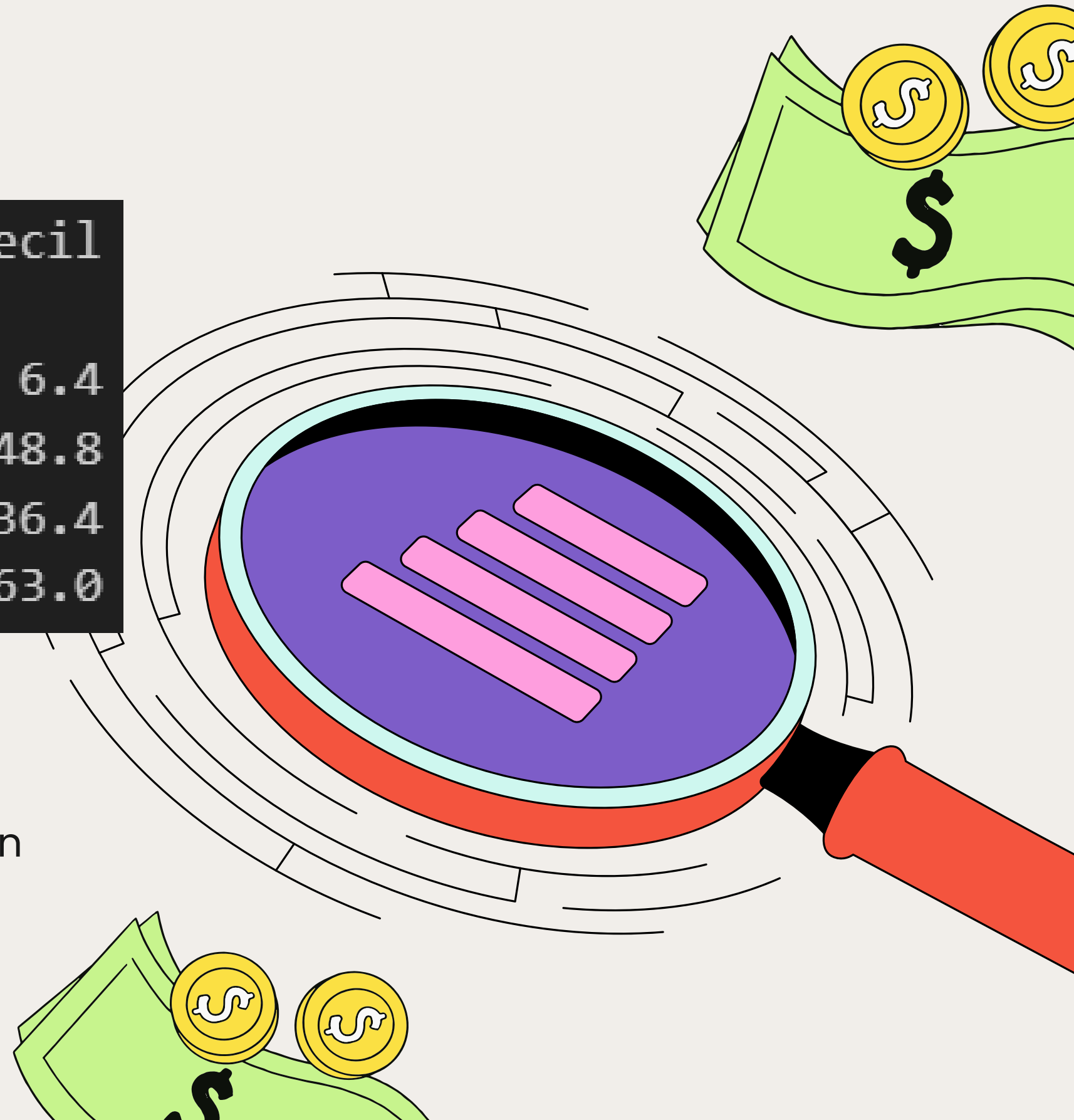
Diskon masih menjadi variabel utama yang mempengaruhi profit

Analisis Lanjutan

profit_bin	Rugi	Untung Besar	Untung Kecil
discount_bin			
3	92.5	1.2	6.4
2	28.5	22.7	48.8
1	19.2	44.4	36.4
0	0.0	37.0	63.0

92.5% transaksi dengan diskon di atas 30% berakhir **Rugi!**
Hanya 1.2% yang Untung Besar, dan 6.4% Untung Kecil.

✓ Kesimpulan: Diskon terlalu tinggi justru menyebabkan kerugian besar. Ini sinyal bahaya untuk strategi pricing.



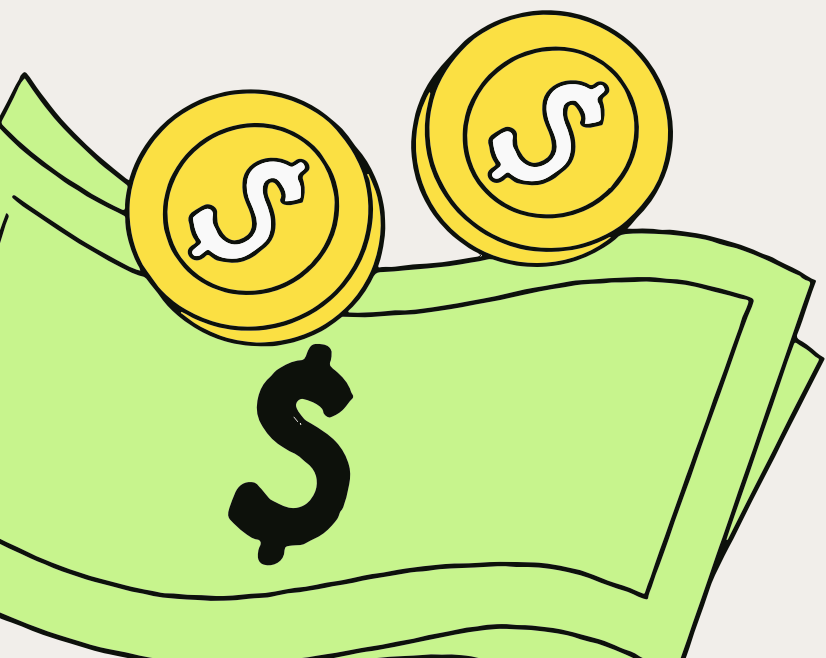
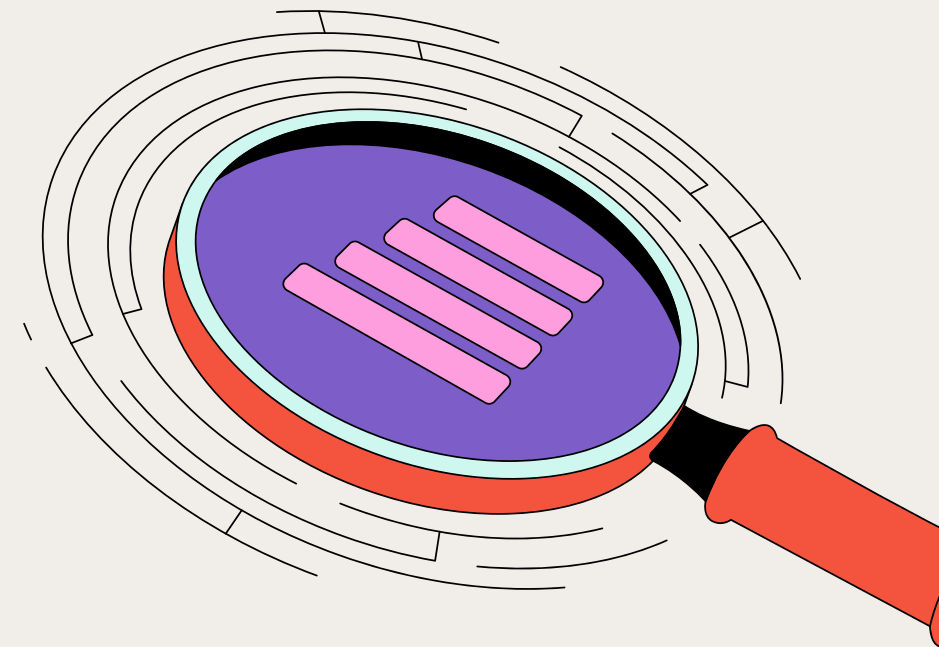


PERLU PENDEKATAN BARU

Optimalisasi Besar
Diskon

**TANPA MENGORBANKAN
MARGIN**

Maksimalkan diskon
kurang dari 30%



Kesimpulan

Accuracy model prediksi profit 85%

Diskon tinggi → meningkatkan beban rugi.

Evaluasi kembali pengaruh diskon terhadap sales

By Data, diskon kurang dari 30% menghasilkan profit yang cukup besar sehingga dapat di optimalkan diskon $<30\%$

